

JAIKE BUSINESS UNIT AI SERVICES

Custom AI-Powered Web Applications & Full-Stack Software Development

At the IJAIKE Business Unit, we provide a variety of AI-powered software development services. Transform your digital presence and internal operations through modern, scalable, and intelligent software. At the IJAIKE Business Unit, we design and develop custom AI-powered web applications, business websites, enterprise portals, and dashboards that combine clean engineering, thoughtful design, and applied AI to deliver fast, reliable, and secure digital products.

Our AI solutions are tailored for startups, enterprises, universities, research organizations, and businesses seeking a dependable technology partner for full-stack software development, from prototype to production deployment. Our AI solutions are intelligently designed and built to improve people's productivity, automate processes, enhance competitiveness and business profitability, and improve customer satisfaction.

Our Full-Stack Software Capabilities at a Glance

- **Business Websites** — professional, responsive, SEO-friendly websites that establish a strong digital presence and convert visitors into customers.
- **AI-Powered Web Applications** — custom web apps that embed intelligent features such as chatbots, recommendation engines, search, and automation directly into the product experience.
- **Enterprise Portals & Dashboards** — secure, role-based portals and real-time dashboards that give teams and clients a single place to manage data, workflows, and reporting.
- **Full-Stack Software Development** — end-to-end engineering across frontend, backend, database, and infrastructure, delivered as a single accountable service.
- **API & Systems Integration** — connecting new and existing applications with CRMs, ERPs, payment gateways, and third-party services via secure, well-documented APIs.
- **Cloud Deployment & DevOps** — reliable, scalable hosting and deployment pipelines that keep applications fast, secure, and easy to maintain.

01 — Our AI Services: What JAIKE Builds

1. Custom Business Websites

Professional, responsive websites for businesses, universities, publishers, and organizations — built for performance, accessibility, and search visibility, with content management systems that make updates easy for non-technical teams.

2. AI-Powered Web Applications

Web applications enriched with embedded intelligence — chatbots, semantic search, recommendation engines, document intelligence, and workflow automation — built directly into the user experience rather than bolted on afterward.

3. Enterprise Portals & Dashboards

Secure, role-based portals for clients, employees, or partners, alongside real-time analytics dashboards that turn raw operational data into clear, actionable insight.

- Client, vendor, and partner portals
- Employee & HR self-service portals
- Real-time analytics & reporting dashboards
- Admin panels & content management consoles

4. Full-Stack Software Development

Complete application development across the entire stack — from database schema and backend APIs to frontend interfaces and deployment — delivered by a single team with end-to-end accountability.

- Frontend engineering (React, Next.js)
- Backend & API development (Node.js, Python)
- Database design & data modeling
- Authentication, authorization & access control

5. API & Third-Party Integrations

Connect your applications with the tools your business already depends on — payment gateways, CRMs, ERPs, email and messaging platforms, calendars, and internal enterprise systems — via secure, well-documented APIs.

6. E-Commerce & SaaS Platforms

Multi-tenant SaaS products and e-commerce storefronts with subscription billing, inventory management, order processing, and customer account management built in from day one.

7. UI/UX Design & Frontend Engineering

Thoughtful interface design paired with clean, component-driven frontend engineering — responsive layouts, accessible design patterns, and consistent design systems across every screen.

- **Wireframing & prototyping:** clickable prototypes that validate flows before development begins
- **Design systems:** reusable component libraries for visual and functional consistency
- **Responsive engineering:** interfaces that work seamlessly across desktop, tablet, and mobile
- **Accessibility:** interfaces built to be usable by the widest possible audience

8. Cloud Deployment & DevOps

Reliable, scalable infrastructure and deployment pipelines that keep applications available, secure, and easy to update.

- CI/CD pipelines for automated testing & deployment
- Containerization & orchestration
- Cloud hosting & environment configuration
- Monitoring, logging & uptime management

9. Database Design & Management

Well-structured relational and non-relational database architectures designed for data integrity, performance, and long-term scalability as your application and user base grow.

10. Performance Optimization & Security

- Application performance audits & load-time optimization
- Security best practices, input validation & vulnerability testing
- Data encryption & secure authentication flows
- Scalability planning for growing traffic and data volumes

11. Maintenance & Ongoing Support

Continued reliability after launch — bug fixes, feature enhancements, dependency updates, and proactive monitoring so applications keep running smoothly as requirements evolve.

12. Embedded AI Features

Optional AI capabilities layered into any web application or portal — conversational assistants, semantic search, document processing, and recommendation systems — reusing JAIKE's Agentic AI stack wherever it adds real value.

13. Consulting & Solution Architecture

Technology strategy, solution architecture, technical feasibility studies, and modernization roadmaps for organizations planning new software or migrating legacy systems.

02 — Technology

Our web applications and enterprise software are engineered on a modern, dependable, and cost-efficient open-source stack spanning frontend, backend, data, and infrastructure:

- React & Next.js
- Node.js & Python
- REST & GraphQL APIs
- MongoDB & PostgreSQL
- Cloud Deployment & Containerization
- CI/CD Automation
- Responsive, Component-Driven UI

Frontend

React	Next.js
TypeScript	Tailwind CSS

Backend

Node.js	Express.js
Python	FastAPI

Databases

MongoDB	PostgreSQL
Redis	SQLite

API & Integration

REST APIs	GraphQL
Webhooks	OAuth 2.0

DevOps & Deployment

Docker	Kubernetes
Git & GitHub	GitHub Actions (CI/CD)
Vercel / Netlify	Render / Railway

Design & Testing

Figma	Storybook
Jest	Playwright

03 — Applications & Deployments

Our web development and full-stack solutions can be deployed for:

- Business & Corporate Websites
- Startup MVPs & SaaS Products
- Client & Partner Portals
- Internal Operations Dashboards
- E-Commerce Storefronts
- University & Research Portals
- Booking & Scheduling Systems
- Content & Publishing Platforms
- Custom CRM & ERP-Adjacent Tools
- Data Visualization & Reporting Systems

04 — Industries We Serve

05 — Why JAIKE Business Unit?

JAIKE Business Unit combines academic rigor with practical software engineering. Every application we build is designed to be scalable, secure, maintainable, and tailored to each client's specific requirements.

Our expertise spans full-stack development, enterprise software, cloud infrastructure, and applied AI — enabling organizations to launch faster, operate more efficiently, and build digital products that grow with the business.

06 -- Call to Action

If your organization is exploring custom AI-powered web applications, business websites, enterprise portals, and dashboards that combine clean engineering, thoughtful design, and applied AI, JAIKE Business Solutions is ready to help you build the next generation of digital products to deliver fast, reliable, and secure solutions.

Contact us at <https://ijaike.org/contacts-us/> or JournalAIKE@Gmail.com to discuss your needs and explore how AI-powered tools can transform your workflows, improve productivity, automate processes, enhance competitiveness and profitability, and enhance customer satisfaction.